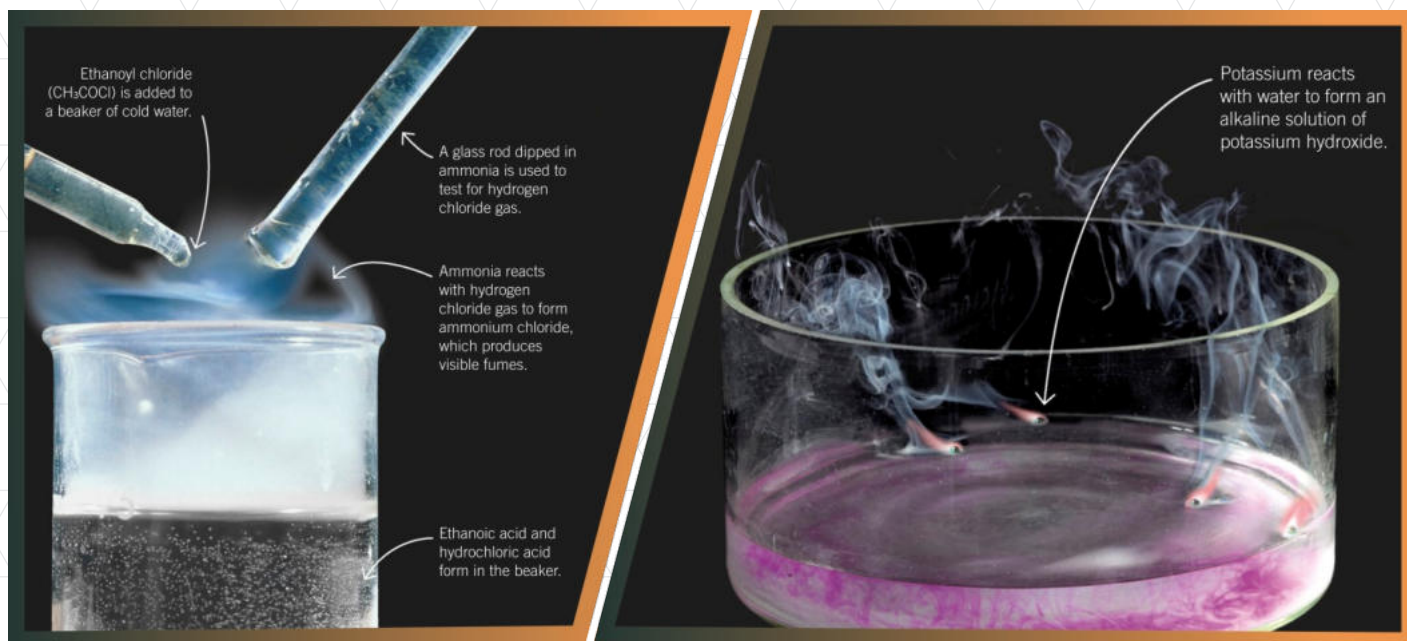




UNIT 02

ACID, BASE AND SALT



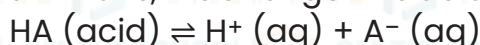
ARRHENIUS THEORY OF ACIDS AND BASES

The Arrhenius theory of acids and bases was proposed by Swedish chemist Svante Arrhenius in 1884. According to this theory, an acid is a substance that dissociates in water to produce hydrogen ions (H^+), and a base is a substance that dissociates in water to produce hydroxide ions (OH^-). This theory is specifically applicable to aqueous solutions.

MAIN POSTULATES OF THE ARRHENIUS THEORY OF ACIDS AND BASES

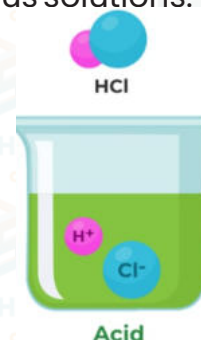
1. ACID

An acid is a substance that ionizes in water to release hydrogen ions (H^+). The higher the concentration of H^+ ions, the stronger the acid.



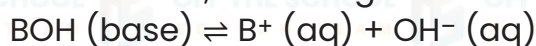
EXAMPLES:

HCl, H_2SO_4 , CH_3COOH .



2. BASE

A base is a substance that ionizes in water to release hydroxide ions (OH^-). The higher the concentration of OH^- ions, the stronger the base.



EXAMPLES:

NaOH, KOH, $Ca(OH)_2$.



3. NEUTRALIZATION

When an acid reacts with a base, they undergo a neutralization reaction, resulting in the formation of water and a salt.



LIMITATIONS

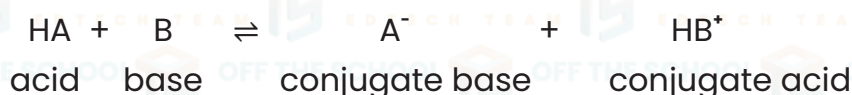
1. Hydrogen ions in water solutions combine with water to form hydronium ions.
2. The Arrhenius theory doesn't account for the basicity of ammonia or the acidity of carbon dioxide and similar compounds.
3. It is only applicable to reactions that take place in aqueous solutions.

BRONSTED LOWRY THEORY OF ACIDS AND BASES

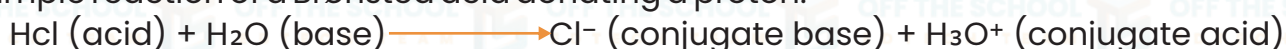
The Bronsted-Lowry theory of acids and bases, proposed by Danish chemist Johannes Brønsted and English chemist Thomas. It defines acids and bases based on their ability to donate or accept protons (H^+ ions) in chemical reactions. According to this theory:

1. BRONSTED ACID

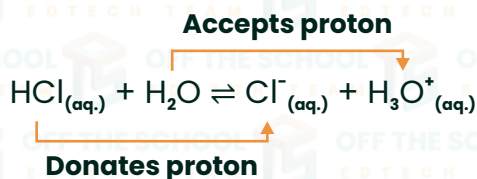
A Bronsted acid is a substance that can donate a proton (H^+ ion) to another species in a chemical reaction. **OR** acid donates its H^+ ion to its conjugate base.



Example reaction of a Brønsted acid donating a proton:



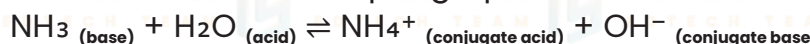
In this reaction, HCl donates a proton to H₂O, forming the chloride ion (Cl⁻) as its conjugate base and a hydronium ion (H₃O⁺) as its conjugate acid.



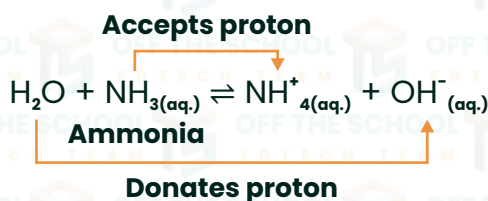
2. BRØNSTED BASE

A Brønsted base is a substance that can accept a proton (H⁺ ion) from another species in a chemical reaction.

Example reaction of a Brønsted base accepting a proton:



In this reaction, NH₃ accepts a proton from H₂O, forming the ammonium ion (NH₄⁺) as its conjugate acid and a hydroxide ion (OH⁻) as its conjugate base.

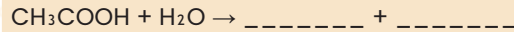
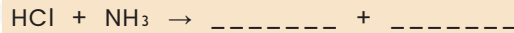


Previous Year's Questions



State three different theories about concepts of acid and base, with example also indicate one limitation of each concept. Also complete conjugated acid and conjugated base of the following reactions.

Acid + Base → Conjugated acid + Conjugated base

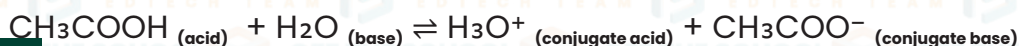


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3. CONJUGATE ACID-BASE PAIRS

In any Brønsted acid-base reaction, there will be the formation of a conjugate acid and a conjugate base.

Example of a conjugate acid-base pair:



LIMITATIONS

1. It does not explain the acidic properties of substances that do not contain hydrogen atoms or H⁺ ions. For example, boron trifluoride (BF₃) and aluminum chloride (AlCl₃) are strong acids, but it does not contain any hydrogen atoms.
2. It does not explain the reactions between acid oxides and basic oxides. For example, carbon dioxide (CO₂) can react with sodium hydroxide (NaOH) to form sodium carbonate (Na₂CO₃), but this reaction does not involve the transfer of protons.

LEWIS THEORY OF ACID-BASE

The Lewis theory of acids and bases was proposed by American chemist Gilbert N. Lewis in 1923. The Lewis theory is based on electron pair donation and acceptance.

Key points of the Lewis theory of acids and bases:

1. LEWIS ACID

A Lewis acid is a species that can accept a pair of electrons. It is an electron-pair acceptor.

2. LEWIS BASE

A Lewis base is a species that can donate a pair of electrons. It is an electron-pair donor.

3. ACID-BASE REACTION

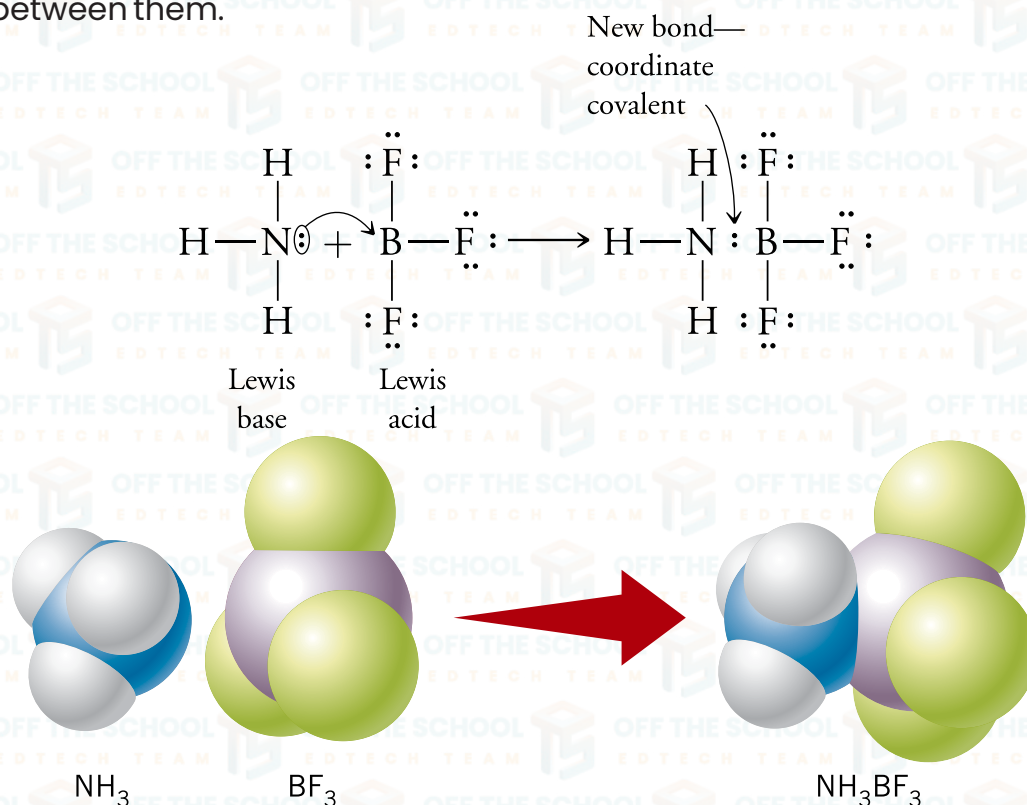
In the Lewis theory, an acid-base reaction involves the donation of an electron pair from the Lewis base to the Lewis acid, forming a coordinate covalent bond between the two species.

EXAMPLE OF A LEWIS ACID-BASE REACTION

Formation of a coordination complex between boron trifluoride (BF_3) and ammonia (NH_3):



In this reaction, ammonia donates a lone pair of electrons to boron trifluoride, forming a new bond between them.



LIMITATIONS

- Limited to electron-pair donation and acceptance:** The Lewis theory focuses on the donation and acceptance of electron pairs. An acid is a substance that accepts an electron pair, and a base is a substance that donates an electron pair. However, it doesn't consider other ways substances can react with each other.
- Doesn't explain reactions involving protons:** The Lewis theory doesn't provide a complete explanation for reactions involving protons (H^+ ions), which are important in acid-base chemistry. It doesn't account for the transfer of protons between substances.
- Doesn't consider solvent effects:** The Lewis theory doesn't consider the role of solvents, such as water, in acid-base reactions. It doesn't explain how the presence of a solvent can influence the behavior of acids and bases.

BALANCING OF NEUTRALIZATION REACTION

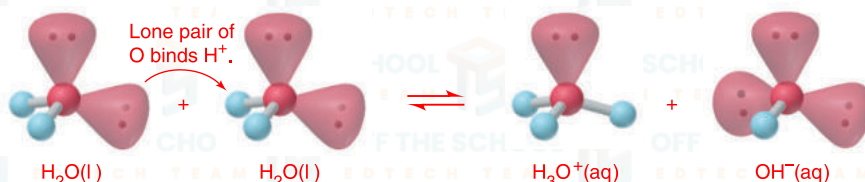
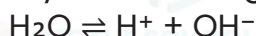
Neutralization reactions occur when an acid reacts with a base in an aqueous solution, resulting in the production of water and a salt. To represent this reaction, a balanced chemical equation is used, ensuring an equal number of atoms and overall charges on both sides.



AUTO-IONIZATION WATER IONIZATION

Water ionization is the process by which water molecules dissociate or break apart into ions in an aqueous solution. In water, a small fraction of water molecules naturally undergo ionization to form hydronium ions (H_3O^+) and hydroxide ions (OH^-).

The ionization of water is represented by the following chemical equation:



The equilibrium constant of (K_c) of water is given by,

$$K_c = \frac{[\text{H}^+][\text{OH}^-]}{[\text{H}_2\text{O}]}$$

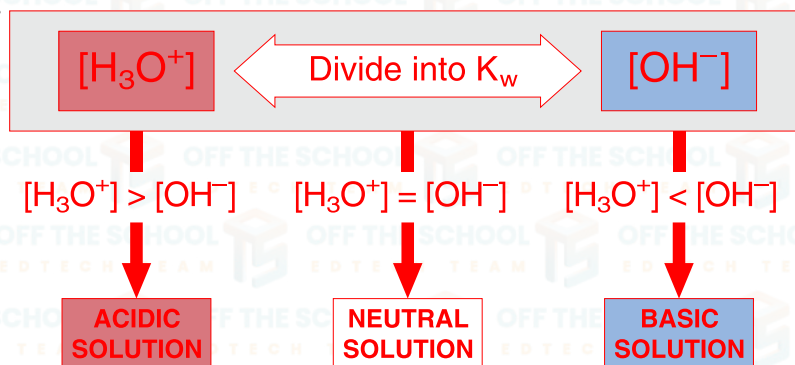
The square brackets represents the molar concentration of species and its units are mole dm. As we know that ionization of water is very small so the concentration is approximately unchanged and consider as constant (K_w) so, the equation will be:

$$K_c[\text{H}_2\text{O}] = [\text{H}^+][\text{OH}^-]$$

$$K_c[\text{H}_2\text{O}] = K_w$$

$$K_w = [\text{H}^+][\text{OH}^-]$$

Where K_w is ionic product constant of water and its value is $1.00 \times 10^{-14} (\text{mol/dm}^3)^2$ at room temperature 25°C .



The relationship between $[\text{H}_3\text{O}^+]$ and $[\text{OH}^-]$ and the relative acidity of solutions.

CONCEPT OF PH AND POH

pH: pH is a measure of the acidity or alkalinity (basicity) of a solution. It quantifies the concentration of hydrogen ions (H^+) present in a solution.

Mathematically, pH is defined as:

$$\text{pH} = -\log[\text{H}^+]$$

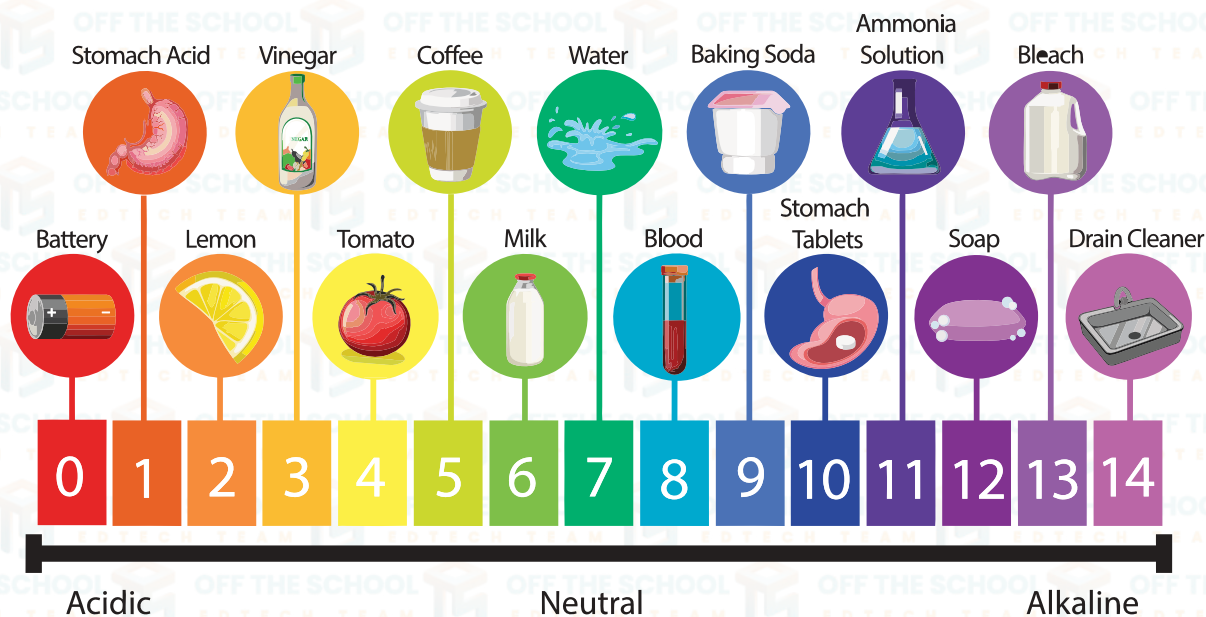
Where $[\text{H}^+]$ represents the concentration of hydrogen ions in moles per liter (mol/L).

pOH: pOH is a measure of the hydroxide ion (OH^-) concentration in a solution. It is the negative logarithm of the hydroxide ion concentration.

Mathematically, pOH is defined as:

$$\text{pOH} = -\log[\text{OH}^-]$$

Where $[\text{OH}^-]$ represents the concentration of hydroxide ions in moles per liter (mol/L).



pH Scale: The Danish biochemist S. P. L. Sørensen devised the pH scale while working on the brewing of beer. The pH scale is a numerical scale used to specify the acidity or basicity (alkalinity) of a solution. It quantifies the concentration of hydrogen ions (H^+) in a solution and ranges from 0 to 14.

NEUTRAL



SALTS

Salt is an ionic compound that contains a cation (from base) and an anion (from acid). It is present in large quantities in seawater, where it is the main mineral constituent.

EXAMPLES: NaCl , CuCl_2 , etc.





1. Salts are produced by the action of acids on metals, metal oxides, metal hydroxide, metal carbonates and metal bicarbonates:



2. Salts are produced by the action of a base with an acid or a metal with a base:



TYPES OF SALTS

1. SIMPLE SALTS

These are salts formed by the direct combination of a metal cation with a non-metal anion.

EXAMPLES: Sodium chloride (NaCl), potassium nitrate (KNO_3), calcium sulphate (CaSO_4).

2. ACID SALTS

Also known as hydrogen salts, acid salts are formed when an acid partially replaces the hydrogen ions with a metal cation.

EXAMPLES: Sodium dihydrogen phosphate (NaH_2PO_4), potassium hydrogen sulphate (KHSO_4).

3. BASIC SALTS

Basic salts are formed when a metal hydroxide reacts with an acid, resulting in the formation of a salt and water.

EXAMPLES: basic lead carbonate ($\text{PbCO}_3 \cdot \text{Pb}(\text{OH})_2$), basic copper sulfate ($\text{CuSO}_4 \cdot 3\text{Cu}(\text{OH})_2$).

USES OF SALTS

1. Food preservation and flavoring in cooking.
2. Water softening to remove hardness in water.
3. Fertilizers for improving crop growth and yield.
4. Pharmaceutical applications in medicine and health.
5. Chemical industry as raw materials and reagents.

CONCEPT OF BUFFER

A buffer is a solution that resists changes in pH when small amounts of acid or base are added to it. It helps to maintain the pH of a solution relatively stable, even with the addition of acidic or basic substances.

TYPES OF BUFFERS

1. ACIDIC BUFFERS

Acidic buffers have a pH below 7 and consist of a weak acid plus its salt with a strong base.

EXAMPLES: A common example is a mixture of acetic acid (CH_3COOH) and sodium acetate (CH_3COONa). This buffer can maintain a pH around 4.75, which is the pK_a value of acetic acid.

2. BASIC BUFFERS

Basic buffers have a pH above 7 and consist of a weak base plus its salt with a strong acid.

EXAMPLES:

An example is a mixture of ammonia (NH_3) and ammonium chloride (NH_4Cl). This buffer system can maintain a pH around 9.25, which corresponds to the pK_a value of the ammonium ion (NH_4^+).

APPLICATIONS OF BUFFER SOLUTION

Buffer solutions are essential in various fields due to their ability to maintain stable pH levels:

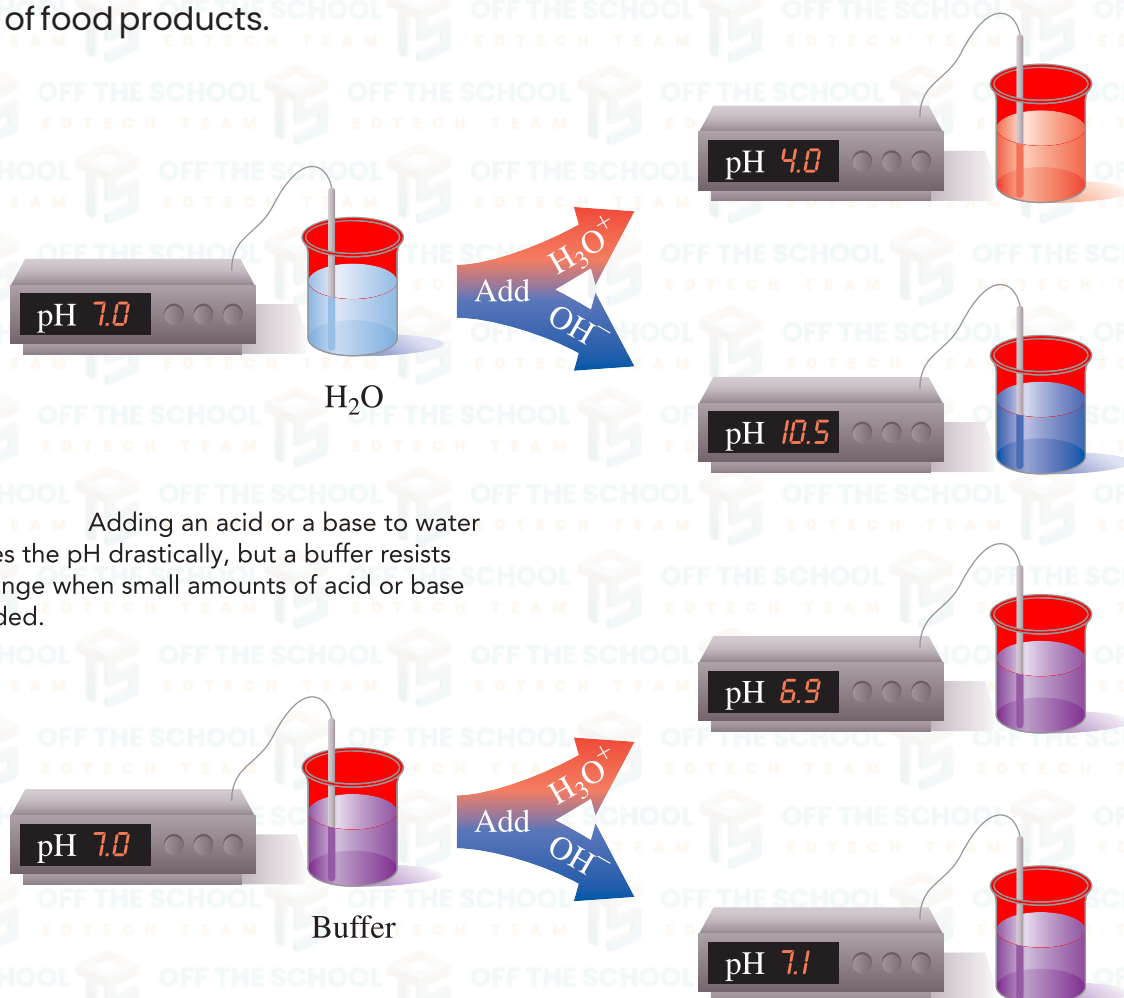
Biochemistry: They regulate blood pH around 7.3 to 7.4 using bicarbonate and carbonic acid buffers.

Industrial Processes: Utilized in fermentation, dyeing, and pharmaceutical manufacturing to ensure optimal pH for reactions.

Agriculture: Used to adjust soil pH to improve crop yields.

Analytical and Pathological Labs: Critical in experiments and tests that require precise pH conditions.

Food Industry: Important for preserving the flavour, appearance, and microbiological stability of food products.



Concept Application Level-01:

Direction for questions 1 to 7: State whether the following statements are true or false.

- 1) Formation of coordinate covalent bond between NH_3 and BF_3 is an example of neutralization reaction, according to Bronsted theory.
- 2) S^{2-} is the conjugate base of HS^- .
- 3) Curd is sour in taste due to the presence of acidic substance.
- 4) K_w has no unit.

Concept Application Level-02:

- 1) Identify the reactant that is Bronsted-Lowry acid and the reactant that is Bronsted-Lowry base in the following reaction:
 - a. $\text{HI} + \text{H}_2\text{O} \rightarrow \text{I}^- + \text{H}_3\text{O}^+$
 - b. $\text{F}^- + \text{H}_2\text{O} \rightleftharpoons \text{HF} + \text{OH}^-$
 - c. $\text{H}_2\text{S} + \text{C}_2\text{H}_5\text{—NH}_2 \rightleftharpoons \text{HS}^- + \text{C}_2\text{H}_5\text{—NH}_3^+$
- 2) Name each of the following acids or bases:
 - a. HCl
 - b. $\text{Ca}(\text{OH})_2$
 - c. HClO_4
 - d. $\text{Sr}(\text{OH})_2$
 - e. H_2SO_3
 - f. HBrO_2
- 3) Name each of the following acids or bases:
 - a. $\text{Al}(\text{OH})_3$
 - b. HBr
 - c. H_2SO_4
 - d. KOH
 - e. HNO_2
 - f. HClO_2
- 4) Write the formula for the conjugate base for each of the following acids:
 - a. HF
 - b. H_2O
 - c. H_2PO_3^-
 - d. HSO_4^-
 - e. HClO_2
- 5) Write the formula for the conjugate base for each of the following acids:
 - a. HCO_3^-
 - b. $\text{CH}_3\text{—NH}_3^+$
 - c. HPO_4^{2-}
 - d. HNO_2
 - e. HBrO
- 6) Write the formula for the conjugate acid for each of the following bases:
 - a. CO_3^{2-}
 - b. H_2O
 - c. H_2PO_4^-
 - d. Br^-
 - e. ClO_4^-

Calculating PH and POH:

- 1) What is the pH of a 0.0235 M HCl solution?
- 2) What is the pOH of a 0.0235 mol/dm³ HCl solution?
- 3) Calculate the pH and pOH of 0.2 M H_2SO_4 ?
- 4) Calculate the pH of 0.1 M KOH ?
- 5) Calculate the pOH of 0.004 M HNO_3 ?
- 6) What is the pH of a 6.50×10^{-3} M KOH solution? (Hint: this is a basic solution – concentration is of OH^-)
- 7) A solution is created by measuring 3.60×10^{-3} moles of NaOH and 5.95×10^{-4} moles of HCl into a container and then water is added until the final volume is 1.00 L. What is the pH of this solution?
- 8) What is the pH of a 6.2×10^{-5} M NaOH solution? (Hint: this is a basic solution – concentration is of OH^-)
- 9) A solution with a H^+ concentration of 1.00×10^{-7} M is said to be neutral. Why?
- 10) Sodium hydroxide is a strong base. What is the pH of a 0.02M sodium hydroxide solution? (Hint: find out pOH and convert this into pH).
- 11) You are presented with a solution that has a pOH of 2.13. What is the pH of this solution?
- 12) What is the pH for a 0.05M solution of hypochlorous acid?
- 13) A sample of gastric juice has a pH of 2.5. What is the hydrogen ion concentration in this secretion?
- 14) Determine the pH at 25°C of a solution prepared by dissolving 0.075 mole of sodium acetate in 1.0 L of 0.25 M acetic acid. (Assume that the addition of sodium acetate does not change the volume of the solution.)

Previous Year's Questions

Find pH, pOH, $[\text{OH}^-]$ and $[\text{H}^+]$ of 2.46×10^{-9} M KOH solution.

[2023–25]